

Difficult meteorological forecasts during a wintertime surface-ozone IOP

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Abstract

ABSTRACT

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1. Introduction

Elevated wintertime surface ozone in Utah’s Uinta Basin occurs when a persistent cold-air pool develops over a snow-covered surface, trapping emissions from the region’s oil and natural gas industry beneath a strong temperature inversion while the high albedo of the snowpack supplies the actinic flux needed to drive photochemistry [?]. Reproducing these episodes in three-dimensional photochemical models has proven persistently difficult: simulations of Uinta Basin winters have tended to underpredict observed ozone, and the field has advanced two broad explanations for the shortfall.

A complicating factor is that the snowpack governing this system is itself spatially diverse, and resolving that diversity is central to prediction [?]. Because the cold pools that concentrate ozone are shallow and confined to the basin floor, coarse operational models can miss high-ozone events entirely: the national Air Quality Model, for instance, runs at roughly 13 km horizontal grid spacing, too coarse to capture the small-scale structure of mountain cold pools [?]. The basin’s position leeward of the Wasatch Mountains produces a suspected precipitation shadow in which subsiding air suppresses snowfall, so snow cover varies sharply across the basin floor; capturing that spatial variation is critical to improved prediction, yet the influence of terrain on snowfall remains poorly constrained, owing in part to the difficulty of high-resolution modeling and to sparse observations across this rural region [?]. These constraints motivate a high-resolution modeling approach.

The first attributes the gap to the emissions inventory. Studies that substituted higher, measurement-derived (“top-down”) volatile organic compound emissions for bottom-up inventory values were able to reproduce the observed 2013 ozone episodes, and concluded that the true emissions must exceed inventoried values [? ?]. Under this view, the limiting factor in earlier simulations was the magnitude of the precursor source rather than the representation of the atmosphere.

A second explanation, which motivates the present work, locates the problem in the meteorology. If a model fails to build and maintain a sufficiently strong inversion, simulated pollutants disperse rather than accumulating beneath the cold-air pool, and ozone is underpredicted regardless of the inventory used. The earliest WRF study of a

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Uinta Basin cold-air pool established a baseline configuration for representing these events [?]. Building on that configuration, ?] sought to improve the simulated meteorology by nudging the model toward observed conditions; because surface observations alone cannot constrain the warm layer aloft that defines an inversion, that approach was later extended using the balloon-borne sonde profiles available for the winter of 2013, the only Uinta Basin winter for which such vertical data exist [?]. Subsequent coupled meteorology–chemistry experiments produced realistic ozone for 2013 under certain chemical mechanisms [?], prompting further work on mechanism sensitivity [?]. That follow-on work, however, relied on a 2019 simulation whose inversion was too weak, limiting its utility [?]; the inadequacy of the 2019 meteorology for representing winter inversions was independently noted in later evaluations [?].

Taken together, this body of work suggests that meteorological fidelity—specifically, the model’s ability to reproduce the strength and persistence of the inversion—is a primary constraint on simulating Uinta Basin winter ozone. Here we [STATE THE AIM: e.g., develop and evaluate a WRF configuration for the winter 2013 cold-air-pool period, using the 2013 sonde profiles to constrain the vertical structure, as a meteorological basis for subsequent photochemical modeling]. [State principal findings.]

2. Materials and Methods

3. Results

3.1. Subsection

4. Discussion

5. Conclusions

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